Task 3: Work with YourTTS

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Abstract

YourTTS, which supports only English, French and Brazilian Portuguese, was used to synthesize an English and a Lithuanian sentence in my voice without any training (zero-shot). In both languages the model copied my pitch well (mean F0 129 Hz vs 123–126 Hz in my recordings), but spoke about 12 % slower, with flatter intonation, a spectrum cut at 8 kHz and extra noise at 4–8 kHz. The English result was [clear]; the Lithuanian one, read with the English setting, had [poor] pronunciation. A dataset of 437 recordings (22.7 min) of EMO speaker M4, emotion neutral, was prepared for future fine-tuning.

1. Model and supported languages

YourTTS is a multilingual, multi-speaker text-to-speech model that can copy a new speaker's voice from a few seconds of speech without training (**zero-shot voice cloning**). It was loaded with the **Coqui TTS** library (`tts_models/multilingual/multi-dataset/your_tts`). The command `print(tts.languages)` returned `['en', 'fr-fr', 'pt-br']`: English, French and Brazilian Portuguese. **Lithuanian is not supported**, so Lithuanian text has to use one of these language settings.

2. Recordings and analysis method

Reference recordings. I recorded on an iPhone the English *"This is a test of the speech synthesis system."* (`reference_en.wav`) and the Lithuanian *"Tai yra šnekos sintezės sistemos testas."* (`reference_lt.wav`). Each served both as the **natural speech** sample and as the **voice reference**. The click at the end of each file (the tap that stopped the recording) was cut off and the stereo files were converted to mono (Table 1).

Recording	Duration	Sampling rate	Format	Channels
reference_en.wav (natural)	3.50 s	48 kHz	WAV, 16-bit PCM	1 (mono)
output_en.wav (synthesized)	3.84 s	16 kHz	WAV, 16-bit PCM	1 (mono)
reference_lt.wav (natural)	3.60 s	48 kHz	WAV, 16-bit PCM	1 (mono)
output_lt.wav (synthesized)	4.14 s	16 kHz	WAV, 16-bit PCM	1 (mono)

Table 1. Parameters of the natural and synthesized recordings.

Analysis. Each signal was loaded with librosa at its original sampling rate. The **waveform** shows amplitude over time, the **magnitude spectrum** (FFT of the whole signal, in dB) shows the energy at each frequency, and the **F0 contour** (pYIN, 60–400 Hz) shows the pitch. I also measured the **speech duration** without silence and the **share of energy at 4–8 kHz**.

3. English zero-shot synthesis

Procedure. The English sentence was synthesized with `tts.tts_to_file(text=..., speaker_wav="reference_en.wav", language="en")`, which produced `output_en.wav` in my voice.

Quality. [2–3 sentences in your own words: Is every word clear? Does it sound like you? Does it sound natural or robotic?]

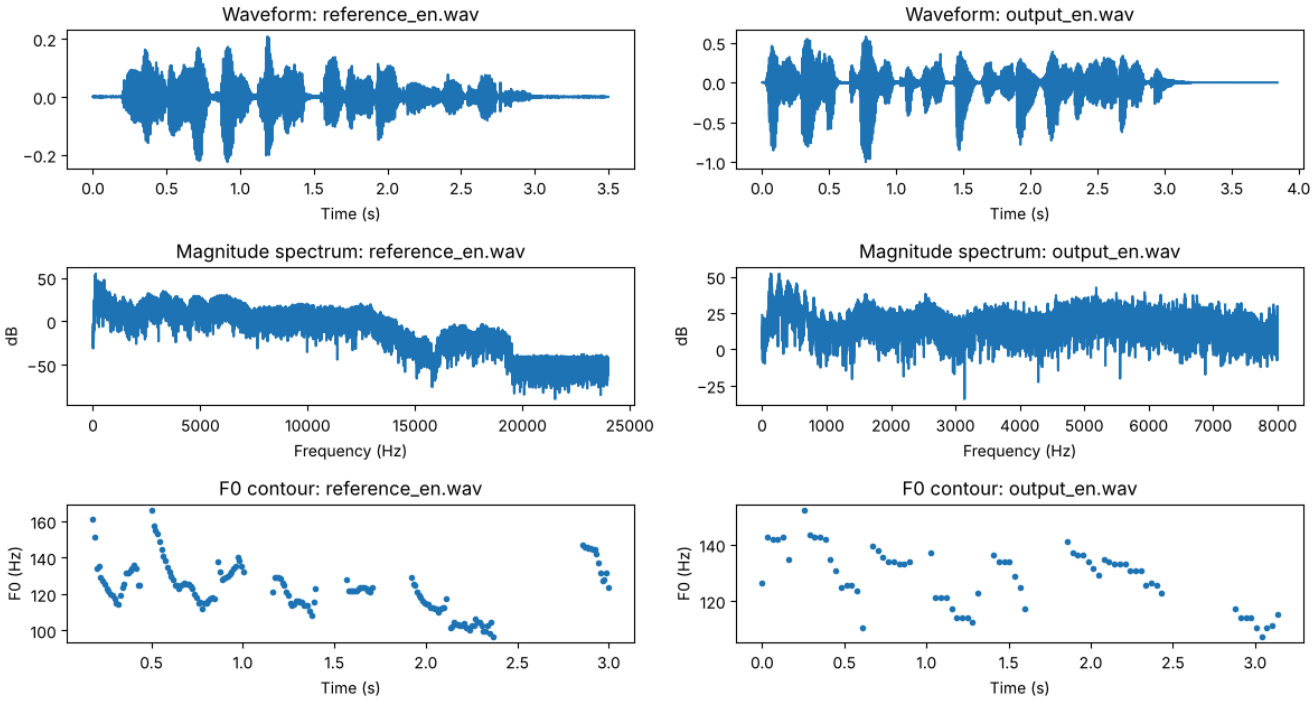


Figure 1. English: waveform, magnitude spectrum and F0 contour of natural (left) and synthesized (right) speech.

	EN natural	EN synthesized	LT natural	LT synthesized
Speech duration (silence removed)	2.83 s	3.17 s	3.10 s	3.46 s
Highest frequency	~13 kHz	8 kHz	~13 kHz	8 kHz

	EN natural	EN synthesized	LT natural	LT synthesized
Energy at 4–8 kHz	5.9 %	18.7 %	4.8 %	7.3 %
Mean F0	123 Hz	129 Hz	126 Hz*	129 Hz
F0 range	97–166 Hz	108–152 Hz	107–149 Hz*	108–163 Hz

*Table 2. Comparison of natural and synthesized speech. * Without F0 errors in the silence after the sentence; the script prints 117 Hz and 60–149 Hz.*

Duration. The synthesized file is longer (3.84 s vs 3.50 s), and so is the speech itself: 3.17 s against 2.83 s, about **12 % slower**. The synthesized speech starts at once, with no silence before it, and ends with 0.67 s of silence.

Spectrum. YourTTS outputs **16 kHz** audio, so its spectrum stops sharply at **8 kHz**, while my recording (48 kHz) has speech energy up to about 13 kHz. Within 0–8 kHz, however, the synthesized speech has **much more energy at 4–8 kHz** (18.7 % vs 5.9 %). The spectrogram shows this energy even during vowels, so it is probably noise added by the model's vocoder rather than real speech sounds.

F0. The mean F0 is **129 Hz against my 123 Hz**, only about one semitone higher, so the model copied my voice pitch well. The synthesized F0 range is smaller (108–152 Hz vs 97–166 Hz), so the intonation is **flatter**. Both contours fall over the sentence, but my recording has a pitch peak on the last word (about 147 Hz at 2.9 s), while the synthesized one ends low (about 110 Hz).

4. Lithuanian zero-shot synthesis

Procedure. Lithuanian is not supported, so the Lithuanian sentence was synthesized with the **English setting** (`language="en"`) and `reference_lt.wav` as the voice reference, giving `output_lt.wav` .

Quality. [2–3 sentences in your own words. Typically the voice still sounds like you, but the text is read with English pronunciation, and letters like "š" and "ė" are skipped or said wrongly (look for "Character … not found" warnings in the console).]

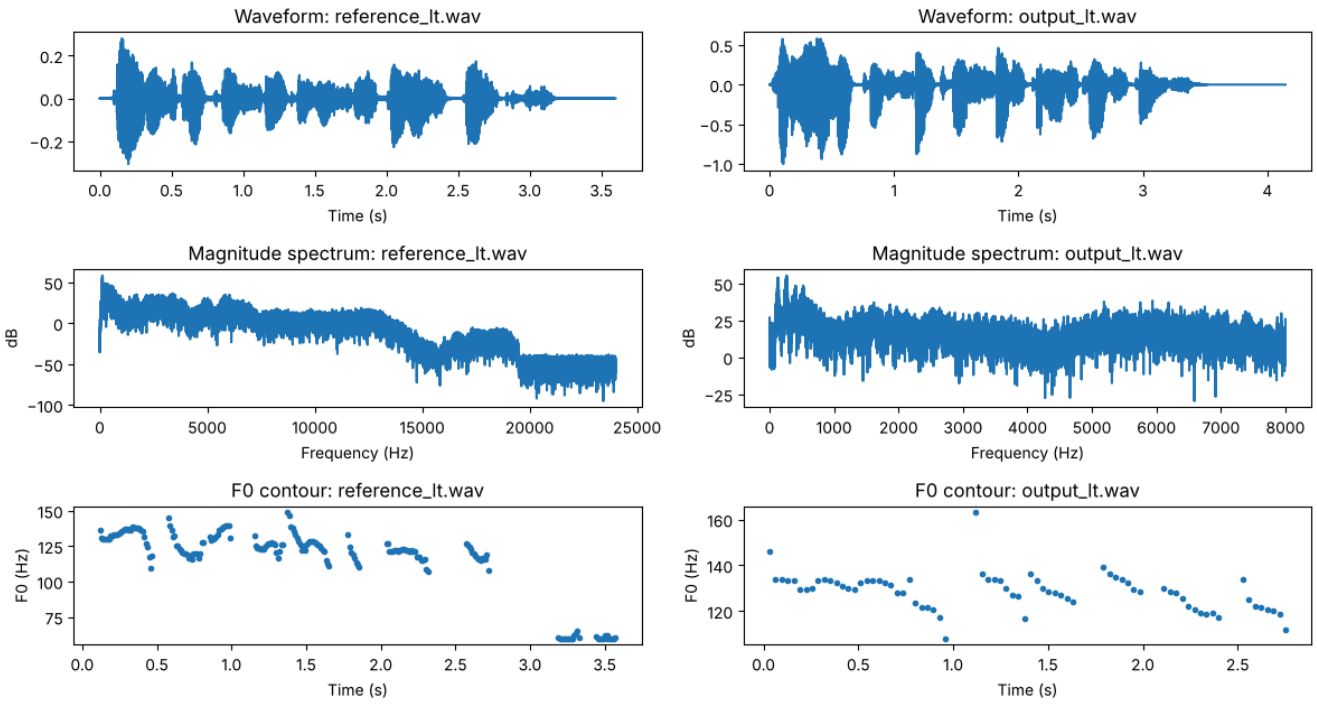


Figure 2. Lithuanian: waveform, magnitude spectrum and F0 contour of natural (left) and synthesized (right) speech.

Duration and spectrum. The synthesized speech lasts 3.46 s against 3.10 s (Table 2), again about **12 % slower**, so the model kept its own speaking rate and did not skip large parts of the text. As in English, the spectrum stops at 8 kHz; the extra energy at 4–8 kHz is smaller here (7.3 % vs 4.8 %).

F0. The mean F0 is close to mine (129 Hz vs 126 Hz). In my F0 plot, the points at 60 Hz after 3.2 s are errors of the pYIN algorithm in the silence after the sentence. The synthesized contour has a regular, repeated shape: almost every word falls from about 135 Hz to 120 Hz. My natural contour is more varied, for example it rises from 120 to 140 Hz at 0.6–1.0 s and peaks at 148 Hz at 1.4 s.

English vs Lithuanian. The measurements are almost the same in both languages, so the voice is copied equally well; the difference is in the pronunciation, which [is much worse in Lithuanian].

5. Lithuanian dataset preparation

The dataset was built from the **Lithuanian Emotional Speech Corpus (EMO)** (Vilnius University, CLARIN-LT, https://hdl.handle.net/20.500.11821/100). Its metadata lists **349,696** transcribed recordings by five speakers (F2, F3, F4 female; M3, M4 male) in five emotions: D – joy, L – sadness, N – surprise, P – anger and Z – neutral. Only **one archive part**, `emo-wavs.zip.021` (312 MB), was downloaded.

Selection. The archive part is one piece of a split zip file, so a normal unzip program cannot open it; a short script (`extract_zip_part.py`) recovered the **437** complete WAV files inside it. Speaker and emotion are coded in each file name (e.g. `M4Z` = speaker M4, emotion Z), and all 437 files belong to speaker **M4** (male) and emotion **Z – neutral**, so all of them were selected. Their transcriptions were taken from `metadata_of_EMO_dataset.csv`.

Split and format. The files were renamed `0001.wav`, `0002.wav`, …, shuffled (seed 42) and split into **90 % training (394)** and **10 % validation (43)**. The metadata files use the Coqui *ljspeech* format, one line per recording: `0001|transcription|transcription`.

Parameter	Value	Parameter	Value
Speaker	M4 (male)	Recordings	437
Emotion	Z – neutral	Total duration	22.7 min
Sampling rate	96 kHz	Minimum duration	0.87 s
Bit depth	24 bit	Maximum duration	7.89 s
Channels	1 (mono)	Average duration	3.12 s
Training set	394	Validation set	43

Table 3. Selected EMO recordings.

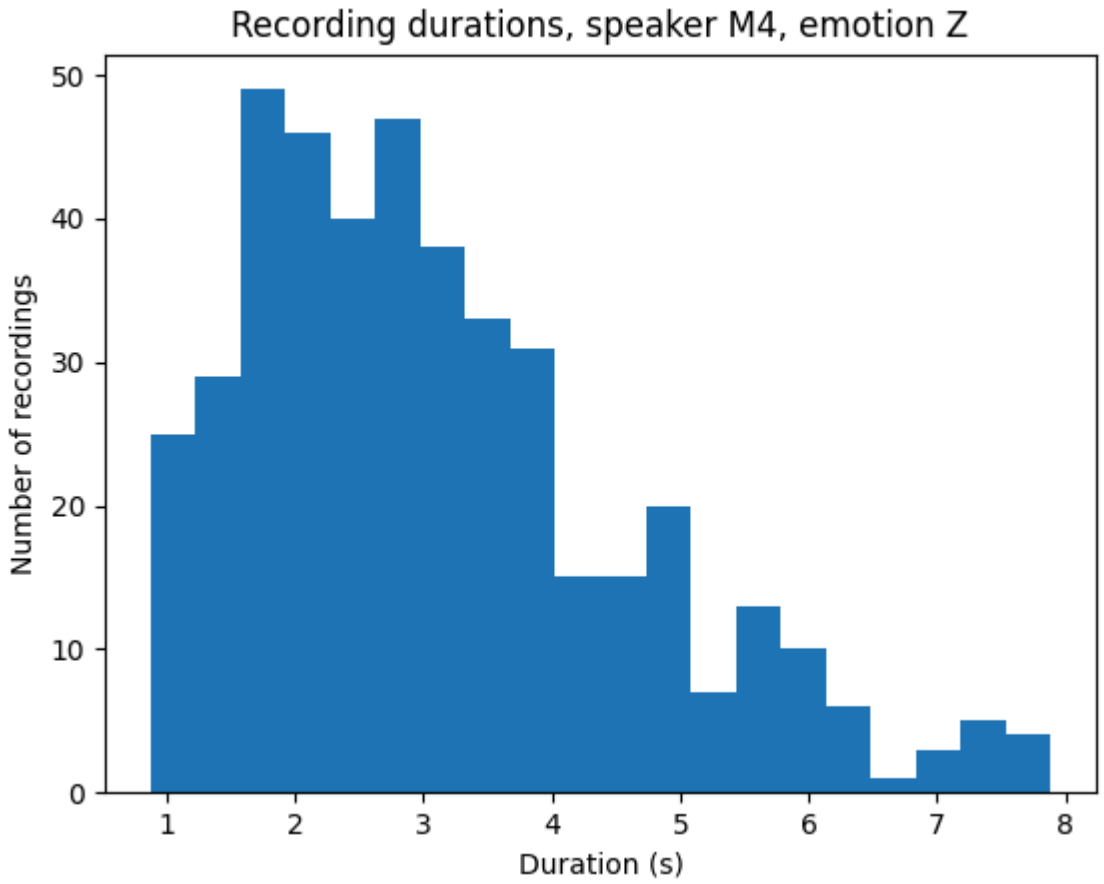


Figure 3. Histogram of recording durations for speaker M4, emotion Z.

Interpretation. Most recordings are short phrases of 1.5–4 s, with the peak around 2 s; fewer last more than 5 s, and only a handful reach 7–8 s. The audio is 96 kHz, 24-bit, much higher quality than the 16 kHz that YourTTS uses, so it would have to be resampled to 16 kHz before fine-tuning.

6. Conclusions

1. **Languages:** the pretrained YourTTS supports English, French and Brazilian Portuguese; Lithuanian is not supported.
2. **Voice:** in both languages the synthesized pitch was close to mine (129 Hz vs 123–126 Hz), so zero-shot voice cloning worked even for the unsupported language.
3. **Differences from natural speech:** the synthesized speech was about 12 % slower, had a smaller F0 range (flatter intonation), no energy above 8 kHz and extra noise at 4–8 kHz.
4. **Lithuanian:** with the English setting the text was read with English pronunciation [and letters like "š" and "ė" were lost], so zero-shot synthesis is not enough for an unsupported language.

5. **Dataset:** for fine-tuning, 437 recordings (22.7 min) of speaker M4, emotion Z (neutral), were prepared from the EMO corpus and split into 394 training and 43 validation recordings.